

ASSIGNMENT REPORT-1

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For The Subject

**Pattern Recognition and Machine Learning
(26AI5707)**



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INTRODUCTION:

In the first part of this experiment, we developed programs to perform essential matrix operations without relying on high-level library functions. The implemented operations include matrix multiplication for matrices of arbitrary dimensions, computation of the dot product between vectors, finding the transpose of a matrix, checking whether a matrix is symmetric, and determining whether a matrix is upper triangular or lower triangular. By implementing these operations from scratch, we gained a deeper understanding of the mathematical principles behind matrix manipulation, which forms the foundation of several machine learning algorithms such as Principal Component Analysis (PCA), Linear Regression, and Neural Networks.

In the second part of the assignment, we focused on probability distributions. We first generated uniformly distributed random variables within a specified interval $[a, b]$ and visualized their distribution using histograms. These uniformly distributed samples were then transformed into Gaussian random variables using the Box–Muller transformation (Marsaglia Polar Method), enabling us to generate normally distributed data with a specified mean (μ) and variance (σ^2). Finally, we plotted histograms of the generated Gaussian samples to verify that the resulting data closely follows the expected bell-shaped normal distribution.

Experiments Performed:

Question 1:

The first experiment involved defining basic operations that are available for matrices and vectors with variable size. In order to do this, we performed the multiplication of matrices by taking the product of corresponding elements in a row and a column of the respective matrices and summing the product values according to the main rule of matrix multiplication. Besides, we carried out the operation of transposition, i.e. we interchanged rows and columns of a matrix. In order to find out if the examined matrix is symmetric, we compared it with its transposed matrix. Moreover, we introduced the methods of determining upper triangular and lower triangular matrices by verifying the values of elements above and below the main diagonal correspondingly. This way, we learned how basic matrix operations are executed within numerical computing libraries and machine learning systems.

$$C_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

where A is an $m \times n$ matrix, B is an $n \times k$ matrix, and the resulting matrix C has dimensions $m \times k$.

the dot product of two vectors is computed as

$$A \cdot B = \sum_{i=1}^n A_i B_i$$

while the transpose of a matrix is represented as

$$(A^T)_{ij} = A_{ji}$$

and is classified as upper triangular if all elements below the principal diagonal are zero, whereas it is lower triangular if all elements above the principal diagonal are zero.

Question 2:

For the second experiment, random numbers were generated uniformly over the interval $[a,b)$. The generated samples were then used to produce a histogram to examine the distribution. As anticipated, the histogram showed equal frequency across the different intervals confirming that the values of the interval were equally likely to occur. There were slight fluctuations in the histogram because of the finite sampling but an increase in the number of samples will lead to a distribution that will be even closer to the theoretical uniform distribution.

Next, the uniformly distributed random numbers are converted into the Gaussian random variables using the Box-Muller transformation. Two independent uniformly distributed variables, i.e. u_1, u_2 were generated in the interval range $(-1,1)$ and value

$$s = u_1^2 + u_2^2$$

was computed. We accepted only those samples satisfying

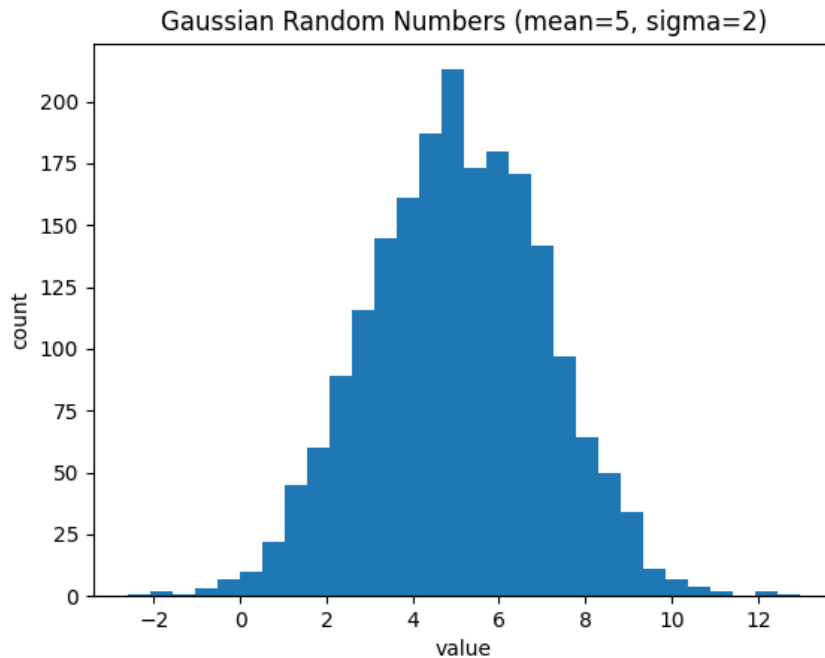
$$0 < s < 1$$

and calculated the scaling factor as

$$k = \sqrt{\frac{-2 \ln(s)}{s}}$$

The histogram of the generated Gaussian samples displayed the characteristic bell-shaped curve, with most samples concentrated around the specified mean and gradually decreasing frequencies toward both extremes. This closely

matched the theoretical normal distribution, confirming the correctness of our implementation.



Inference

This lab assignment has allowed us to deepen our knowledge of essential theory involved in pattern recognition and machine learning. From this task, we could see that different programming libraries provide similar functions based on basic mathematical operations. The process of working with basic matrix computations has allowed us to gain insight into the process of working with matrices and their significance in machine learning algorithms. The experiments carried out on generation of random numbers provided us with an opportunity to discover what the difference between uniform and Gaussian distributions is. By drawing graphs, we had a chance to/were able to see how uniformly distributed data differs from normally distributed data. After the Box–Muller transformation was made, we found it much easier to comprehend how to apply mathematical formulas for generating Gaussian random variables from uniformly distributed random numbers.

